

Xana Plus Super App — Product Requirements Document

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1. Purpose, vision and scope

Xana Plus is one app where a Kenyan household buys medicine, sees a clinician, books a lab test, claims SHA or private insurance, buys household essentials in bulk, and manages its health day to day. This PRD specifies the full vision for a build team. Phasing and budget are out of scope and will be set separately.

Vision. Become the default first stop in Kenya for health, wellness, nutrition and household health spending. The pharmacy network is the trust anchor; the app extends it to every point of care.

Product principles

- One identity, one wallet, one health record across every module.
- Works on low-end Android, on 3G, and offline for core flows.
- M-Pesa first. Card and insurance are secondary rails.
- Every clinical action has a licensed human accountable for it.
- Consent before data. The user controls who sees their record.
- Physical stores are part of the product: order online, collect, consult or test in-store.

In scope: pharmacy retail and delivery, clinics and telehealth, lab and diagnostics, SHA and private insurance, Xana Wholesale bulk buying, Xana Plus paid membership, wellness, nutrition and community, the personal health record, payments, and the supporting platform.

Out of scope for this document: inpatient hospital services, emergency ambulance dispatch, and the Afyanalytics research products themselves (only the consented data hand-off is specified here).

2. Users and personas

The app serves five consumer personas and four internal or partner roles. Personas below are working assumptions for design; they should be validated with user research before build.

Persona	Who	Primary jobs in the app
Household health manager	Parent, usually a mother, managing care for children and elderly relatives	Family profiles, refills, child consults, vaccination reminders, bulk buying
Chronic patient	Adult with a long-term condition needing monthly medicine	Auto-refill, adherence reminders, remote monitoring, SHA or insurer claims
Young urban professional	22–35, smartphone native, Nairobi or Mombasa	Quick teleconsult, wellness, fitness and nutrition, community

Persona	Who	Primary jobs in the app
Peri-urban / rural user	Lower bandwidth, feature phone in household, cash-constrained	USSD / WhatsApp access, pickup at store, pay in instalments
Small business buyer	Duka owner, school, clinic or employer buying in bulk	Xana Wholesale catalogue, credit terms, invoices with eTIMS

Internal / partner role	Uses the app for
Pharmacist (Xana store)	Verifying prescriptions, dispensing, counselling, order picking
Clinician (doctor, clinical officer, nurse)	Teleconsults, e-prescribing, lab orders, follow-ups
Lab partner / phlebotomist	Sample collection, results upload
Rider / delivery partner	Pickup and last-mile delivery with proof of delivery

All consumer personas need Swahili and English from day one. Other languages are an open question (section 17).

3. App structure

The app is a shell with five bottom tabs, a set of service modules, and shared platform services that every module must use. No module may build its own login, payments, notifications or record store.

flowchart TD

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A[Xana Plus shell] --> B[Home]
A --> C[Shop]
A --> D[Care]
A --> E[Me & Family]
A --> F[Community]
C --> C1[Pharmacy]
C --> C2[Wholesale]
D --> D1[Telehealth & clinics]
D --> D2[Labs]
D --> D3[Insurance & SHA]
E --> E1[Health record]
E --> E2[Wellness & nutrition]
E --> E3[Membership & wallet]

```

The shell sits on shared services: identity and family profiles, wallet and payments, health record and consent, notifications, search, support chat, store locator and analytics.

Tab behaviour

- **Home:** personalised feed. Due refills, upcoming appointments, pending lab results, claim status, member offers, one health tip. A universal search bar searches products, conditions, services and stores.
- **Shop:** one cart across pharmacy and wholesale items. Prescription items are flagged and gated.
- **Care:** start a consult, book a clinic or lab, check coverage, file a claim.
- **Me & Family:** profiles for self and dependants, record, medications, wellness goals, membership, wallet, settings.
- **Community:** moderated groups, expert Q&A, challenges.

Cross-module journeys the build must support end to end

1. Teleconsult → e-prescription → coverage check → pay → delivery or pickup → adherence reminders.
2. Clinician orders lab → home collection booked → result lands in record → clinician follow-up auto-scheduled.
3. Chronic refill due → auto-order → SHA or insurer pre-authorisation → dispatch.
4. Member scans a product in store → sees member price → adds to app cart for delivery.

Alternative access channels: WhatsApp and USSD front-ends cover refills, order status, appointment booking and payment. They call the same backend APIs as the app. A lightweight PWA covers users who will not install an app.

4. Pharmacy: retail, refills and delivery

The pharmacy module sells OTC, prescription and front-of-store products from live Xana store stock, with pharmacist verification on every prescription line. Stock, pricing and orders are mastered in Business Central; the app never holds its own inventory truth.

Catalogue and search

- Product pages: generic and brand name, strength, pack size, manufacturer, price, member price, stock by nearby store, OTC or Rx badge.
- Generic substitution suggestions shown with price difference, subject to pharmacist approval on Rx items.
- Search by product, active ingredient, symptom or condition. Symptom searches show OTC options plus a "talk to a pharmacist" prompt.
- Barcode scan in store to see member price and add to cart.

Prescriptions

- Three entry routes: e-prescription from a Xana clinician, photo upload of a paper script, or e-prescription from an external prescriber (if a national e-Rx standard is available; see section 17).
- Every Rx line enters a pharmacist queue. Pharmacist checks validity, interactions, dose and allergies against the user's record, then approves, queries or rejects.
- Controlled or restricted medicines follow extra rules set by the compliance team; the app must support blocking them from delivery entirely.
- Prescription history and remaining repeats shown per family member.

Refills and adherence

- Auto-refill schedule per medication, with a reminder 5 days before run-out (configurable).
- One-tap reorder. Chronic refills can be bundled into one monthly delivery.
- Dose reminders with "taken / skipped" logging; adherence summary visible to the user and, with consent, their clinician.

Fulfilment

- Options: express delivery, scheduled delivery, click-and-collect, and pickup lockers if deployed.
- Order routing picks the nearest store with full stock; split orders only with user consent.
- Rider app: pickup scan, live tracking for the user, cold-chain flag for items like insulin, OTP or photo proof of delivery.
- Order states: placed → pharmacist review → paid → picking → ready → out for delivery → delivered / collected, plus cancelled and returned.

Acceptance criteria (sample)

- No Rx item can be dispatched without a logged pharmacist approval ID.
- Stock shown in the app is no more than 15 minutes behind Business Central (target to be confirmed with the Business Central integration partner).
- A refill reorder takes no more than 3 taps from the reminder notification.

5. Clinics and telehealth

Users can consult a licensed clinician by chat, voice or video, or book an in-person visit at a Xana or partner clinic, and every consult writes to the same health record. Whether Xana employs clinicians, contracts them, or partners with an existing telehealth provider is an open decision (section 17); the spec must work for all three.

Consult types

Type	Channel	Typical use
Pharmacist chat	Text	Minor ailments, OTC advice, medication questions
On-demand GP consult	Chat, voice, video	Acute illness, triage, prescriptions
Scheduled specialist	Video or in-person	Paediatrics, gynaecology, chronic disease review, mental health
In-store clinic visit	In person	Vitals, vaccinations, minor procedures, screening
Chronic care programme	Mixed	Scheduled reviews plus remote monitoring

Flow

1. Structured intake: who is the patient (self or dependant), symptoms, duration, current medicines, allergies.
2. Rule-based red-flag screen. Emergency symptoms route immediately to an emergency message with the nearest facility and a call button; the app does not continue to a routine consult.
3. Queue or booking, with estimated wait time and price (or "covered" if the user's plan pays).
4. Consult with record access (with consent), in-call notes, e-prescription, lab orders, sick note and referral letter.
5. Post-consult summary in plain language, follow-up booked, prescription pushed to the pharmacy cart.

Clinician console (web)

- Queue, schedule, patient record view, templated notes, e-prescribing with interaction checks, lab ordering, referral directory.
- Licence number and specialty stored and shown on every prescription and note.
- Audit log of every record access.

Technical requirements

- Video must degrade gracefully to voice, then to chat, on poor connections.
- Consults are not recorded by default. If recording is ever enabled, it requires explicit consent per session.
- Clinic booking syncs with the clinic's own scheduling system where one exists.

6. Lab tests and diagnostics

Users can book lab tests ordered by a clinician or chosen directly, with home collection or in-store sampling, and receive results in their record with a clinician explanation. Labs are assumed to be accredited partners, not Xana-owned; the integration must support multiple labs.

Features

- Test catalogue: single tests and packages (for example diabetes check, antenatal panel, executive screen), with price, preparation instructions and turnaround time.
- Self-ordered tests allowed only from an approved list set by the clinical lead. Other tests require a clinician order.
- Booking: home collection slot, store sampling slot, or walk-in at partner lab. Phlebotomist assignment and tracking like a delivery.
- Sample chain of custody: barcode label scanned at collection, handover and lab receipt.
- Results: structured values where the lab can send them, PDF otherwise. Reference ranges shown. Abnormal results trigger a clinician review task before or alongside release to the user, per clinical lead policy.
- Trends over time for repeat tests (for example HbA1c, cholesterol).
- Point-of-care tests done in store (glucose, blood pressure, rapid tests) are logged to the record by the pharmacist.

Integration

- Preferred: HL7 FHIR ServiceRequest / DiagnosticReport exchange with each lab's LIS. Fallback: secure portal upload by the lab.
- Result delivery time tracked per lab for partner SLA management.

7. Insurance and SHA claims

Users link their SHA membership and any private or employer cover once, then see what is covered at checkout and let Xana file the claim. Integration specifics are the highest-risk area of this PRD: SHA's current technical interfaces and accreditation rules for pharmacies, telehealth and labs must be confirmed before design is finalised.

Features

- Link cover: SHA member number and private policies (insurer, member number, principal and dependants). Eligibility check on link and before each covered transaction.
- Coverage at checkout: for each line, show covered, partially covered (with co-pay) or not covered. User pays only the co-pay.
- Pre-authorisation requests generated automatically for items that need them, with status tracking.
- Claims: Xana files on the user's behalf wherever Xana is the accredited provider. Users can also upload receipts for reimbursement claims to their own insurer.
- Claim tracker: submitted, under review, approved, paid, rejected (with reason and appeal path).
- Benefits view: remaining limits per benefit category where the insurer exposes them.
- Employer schemes: corporate admin portal to enrol staff, set benefit rules and see aggregate (never individual clinical) usage.

Integration approach

- SHA: integrate through whatever official API or claims system SHA provides to accredited facilities. I do not have verified, current details of SHA's interfaces; the team should obtain them directly from SHA before estimating this work.
- Private insurers: many Kenyan insurers route claims through third-party claims-switch platforms. Candidates are thought to include SMART and Slade360, but which insurers use which, and on what terms, must be verified.
- Design an adapter layer so each payer is a plug-in with its own eligibility, pre-auth and claim endpoints.

Acceptance criteria (sample)

- Coverage status per line is shown before the user confirms payment.
- Every claim stores the underlying prescription, invoice and eTIMS receipt reference for audit.

8. Xana Wholesale: bulk buying

Xana Wholesale sells health, wellness, baby, personal care and household staples in bulk packs at warehouse prices, Costco-style, to members and business accounts. It shares the cart, wallet and delivery network with the pharmacy but has its own catalogue, pricing and fulfilment rules.

Who can buy

Account	Access	Extras
Xana Plus member (consumer)	Full bulk catalogue plus member specials	Member pricing, free delivery thresholds
Non-member consumer	Full bulk catalogue	Prompt to join
Business account	Full catalogue plus trade packs	Credit terms, tax invoices, multiple users, purchase approvals

Features

- Bulk and case-pack SKUs with unit price shown per item so savings are visible.
- Household "stock-up lists" and subscribe-and-save on recurring items (nappies, formula, sanitary products, vitamins).
- Group buying: a user starts a group order (chama, estate, office), others join, and a combined order ships to one address at a tier price once a threshold is met. Each participant pays their share via M-Pesa.
- Business accounts: KRA PIN capture, eTIMS-compliant invoices, net-terms credit subject to credit checks, order approval workflow, reorder from history, bulk CSV upload of orders.
- Prescription-only medicines are never sold through Wholesale to consumers. Any B2B supply of pharmaceuticals to licensed buyers (for example clinics or other pharmacies) needs a separate licence check and is flagged for compliance review in section 17.
- Fulfilment from a warehouse or distribution centre, with scheduled delivery windows and larger vehicles for big orders.

Acceptance criteria (sample)

- Group-buy orders auto-refund every participant if the threshold is not met by the deadline.
- Business invoices are generated and transmitted through eTIMS at the point of sale.

9. Xana Plus membership

Xana Plus is a paid subscription that bundles savings and care, so a member family gets more value than the fee every month. Tier names and benefits below are a proposed structure; prices are deliberately left blank for the commercial team to set.

Benefit	Free	Plus Individual	Plus Family (up to 6)	Plus Business

Benefit	Free	Plus Individual	Plus Family (up to 6)	Plus Business
Pharmacy member pricing	No	Yes	Yes	Yes
Wholesale access	Limited	Yes	Yes	Yes, with trade packs
Free delivery	Above threshold	Lower threshold	Lower threshold	Scheduled bulk delivery
Pharmacist chat	Yes	Yes	Yes	Yes
GP teleconsults	Pay per use	Included quota	Shared family quota	Per-employee quota
Annual screening package	No	Discounted	Discounted per member	Staff screening days
Chronic care programme	Pay per use	Discounted	Discounted	Optional add-on
Priority support	No	Yes	Yes	Account manager
Price (KES/month)	0	TBD	TBD	TBD

Billing

- Monthly or annual billing. M-Pesa recurring payment if available to Xana (verify current Safaricom recurring-payment options), otherwise reminder-and-pay, card auto-charge or wallet debit.
- Grace period and pause option; benefits suspend, not cancel, on failed payment.
- Member savings tracker on Home: "You saved KES X this year" to reinforce renewal.
- Employer-sponsored and insurer-bundled memberships supported via the corporate portal.

Rules

- Membership benefits never override clinical rules (for example, a teleconsult quota does not guarantee a prescription).
- Quotas are enforced server-side and visible to the user.

10. Wellness, nutrition and community

This module is what brings users back weekly when they are not sick: tracking, locally relevant nutrition guidance, and moderated community. All content must be reviewed by a qualified professional before publishing, and the module must never present itself as diagnosis.

Wellness tracking

- Manual logging plus sync with phone health data (Google Health Connect, Apple Health) and common wearables, via their official SDKs.
- Trackers: steps, weight, blood pressure, glucose, sleep, water, mood, menstrual cycle, pregnancy week, child growth (with WHO growth charts).
- Readings from in-store or home devices flow into the record and can alert the care team for chronic-programme patients against thresholds the clinician sets.

Nutrition

- Kenyan food database (ugali, sukuma wiki, githeri, chapati, nyama choma and so on) with portion sizes in local measures. The source for nutrient values must be a verified food composition table; the team should confirm which Kenyan or East African table to license.
- Meal logging by photo, search or quick-add.
- Condition-aware meal plans (diabetes, hypertension, pregnancy, child weaning) authored by registered nutritionists, with budget-level variants.
- Shoppable plans: ingredients or supplements link to the Shop where relevant.
- Safety rule: no calorie targets or weight-loss plans are shown to users under 18, and the app must include screening and safe-messaging rules for disordered-eating risk, designed with a clinician.

Programmes and challenges

- Structured programmes (for example 12-week blood pressure, pregnancy journey, quit smoking) with content, check-ins and optional clinician touchpoints.
- Challenges (step challenges, water, family activity) with points redeemable as wallet credit.

Community

- Topic and condition groups (new mothers, diabetes, fitness, caregivers), plus local groups by estate or town.
- Expert Q&A sessions and AMAs with Xana clinicians and nutritionists; expert answers are badged.
- Moderation: automated filtering plus human moderators; medical misinformation and product selling by users are removed; crisis language routes to support resources.
- Privacy: pseudonymous display names by default; nothing from the health record is ever shared to community automatically.

Content hub

- Articles, short videos and audio in Swahili and English, tagged by condition, life stage and season (for example malaria season, back-to-school).

10A. Flourishing and social layer

This layer gives users a reason to open Xana weekly or daily when they are not sick, by organising the app around whole-person flourishing and private, trusted communities. Health is low-frequency; family, community, faith and money are not, so this layer is the product's engagement engine and its main differentiator.

Organising framework

The app's wellbeing experience is organised around flourishing domains adapted from Tyler VanderWeele's Human Flourishing Program at Harvard, extended with shalom's wholeness with God and creation.

Domain	Example features
Physical health	Sections 4–7 and 10
Mental and emotional health	Counselling, peer groups, mood journal, grief support
Close relationships	Care Circles, caregiver support, family programmes
Financial stability	Medical harambee, health chamas, financial wellness
Meaning, purpose and faith	Opt-in faith layer, volunteering, giving
Character and habits	Challenges, group activity, healthy-behaviour rewards
Creation and environment	Farm-to-table marketplace, local environmental alerts, kitchen gardens

A periodic, private self-reflection across these domains (a "flourishing check-in") suggests programmes and resources. It is never shown to others, never ranked, and never used for insurance pricing.

Care Circles (core feature)

- Private groups of 2–50 people: family, chama, church small group, estate, office team.
- Circle chat (text, voice notes, photos), shared reminders and tasks ("Mum's BP meds due"), check-ins ("How are you today?" with one-tap replies), shared goals and group challenges.
- Care roles: a circle can designate a caregiver for a member, with that member's explicit, revocable consent to share specific items (for example medication adherence, appointment dates). Nothing from the health record reaches a circle without such consent.
- Circle admins control membership; users can leave any circle at any time.

Diaspora care

- A user abroad adds a relative in Kenya and pays for refills, consults, lab tests, membership or wholesale orders on their behalf, by international card or remittance partners (partners to be selected).
- The relative receives care through a Xana store, delivery or home visit, even without a smartphone (USSD or SMS confirmation).
- The payer receives proof of service (delivered, consult completed). Clinical details are shared only with the relative's consent.

Care gifting

- Send a voucher for a consult, lab test, screening package, membership or pharmacy credit to any phone number, with a personal message. Themes for birthdays, new babies, weddings, and get-well.
- Vouchers are redeemable only for the stated service category and expire under published terms.

Medical harambee

- A user raises funds for a verified medical bill. Xana or a partner facility confirms the bill amount before the campaign goes live.
- Donors pay by M-Pesa or card; funds are paid directly to the provider, not to the individual, with a public progress bar and a confirmation when the bill is settled.
- Campaign pages show only what the patient or guardian chooses to share.
- Legal structure (for example whether Xana acts as a collection agent or uses a licensed partner) must be confirmed by counsel before build. Existing Kenyan crowdfunding platforms should be reviewed as partners or competitors.

Health chamas

- Group savings pots earmarked for members' health costs, operated through a licensed financial partner, with contributions by M-Pesa and transparent balances for all members.

Opt-in faith layer

- Off by default; the user chooses a faith tradition or none. No faith content appears unless enabled.
- Features: daily devotional or reflection with a health or wellbeing thought, prayer circles (a Care Circle type), gratitude journal, Sabbath and rest reminders.
- Churches, mosques and faith organisations can run official accounts and circles for their members.
- Content partners are vetted and content is reviewed for accuracy and for respect across traditions.

Volunteering and giving

- Listings for community health days, blood drives and service opportunities from verified organisations.
- "Sponsor a family's care": donate care vouchers to families identified by partner churches, NGOs or community health workers.

Creation and environment

- Farm-to-table marketplace: produce from vetted local farmers, linked to nutrition meal plans and fulfilled through Wholesale logistics.
- Local alerts for air quality, water safety and seasonal disease risk, only from verified data sources (sources to be identified).
- Kitchen-garden and urban farming guides.

Healthy-behaviour rewards

- Points for verified activities (completed programmes, screenings, adherence, challenges), redeemable as pharmacy or wholesale credit.
- Designed so insurers and employers can fund rewards for their members, following the model of programmes such as Discovery Vitality in South Africa.
- Rewards never penalise users for health status, only reward effort.

Mini-programs and official accounts (platform)

- Approved third parties (gyms, counsellors, schools, churches, farms, employers, NGOs) build mini-programs inside Xana using Xana identity, payments, circles and notifications through a partner SDK.
- Each partner passes a vetting process; health-related partners also pass licence checks. Mini-programs cannot read the health record without explicit per-partner user consent.
- Official accounts let verified organisations publish content and message followers, with rate limits and opt-out.
- QR codes throughout: scan to join a circle, follow an account, pay, check in at a clinic or event.

Expert short video

- Short-form video published only by verified clinicians, nutritionists, counsellors and partner organisations. No user-generated health advice videos.

Swahili voice assistant

- Voice navigation in Swahili and English for low-literacy users: reorder medicine, book appointments, hear results explained in plain language.
- Decision support only, clearly labelled, logged, and clinically validated before release (see section 14).

Social design rules

- No public open feed at launch; social happens in circles and official accounts.
- No public leaderboards for weight, appearance or body measurements. Challenge rankings, if any, are within a circle and opt-in.
- Health misinformation is removed; verified expert answers are badged.
- Users under 18 get restricted social features designed with child-safety guidance.
- Crisis language anywhere in circles or chat routes to support resources.

Acceptance criteria (sample)

- No health record data appears in any circle, campaign or mini-program without a logged, specific consent.
- Harambee funds can only be disbursed to a verified provider account.
- The faith layer shows no content to a user who has not enabled it.

11. Personal health record, consent and data layer

Every user and dependant has one longitudinal health record, owned and controlled by the user, that every module reads from and writes to. It is the product's core asset and must be built to FHIR R4 from the start.

Record contents

- Demographics, family links, allergies, conditions, medications (current and past), prescriptions, consult notes and summaries, lab results, vitals and tracker data, immunisations, documents (uploaded scripts, discharge summaries).

Family and delegation

- A guardian manages records for children. Adult dependants (for example an elderly parent) grant delegated access that they can revoke.
- Access for a child's record transfers to the child at age 18, with the guardian notified.

Consent model

Consent	Default	Granularity
Xana clinicians and pharmacists access record for care	Asked at first consult	Per care episode, revocable
Share with external provider	Off	Per provider, time-limited link or QR
Insurer / SHA receives claim data	Required for claims	Only data needed for that claim
De-identified data for research (e.g. Afyanalytics)	Off	Separate, explicit opt-in; revocable for future use
Marketing personalisation from health data	Off	Explicit opt-in

Data layer requirements

- Every read and write to the record is audit-logged with who, when and why.
- Any research data feed is de-identified to a documented standard and passes through a governance review before release. The legal basis, de-identification method and data-sharing agreement with Afyanalytics must be approved by counsel and the data protection officer.
- Record export for the user in human-readable PDF and FHIR JSON.
- Interoperability with national health information exchange or client registry if one is available to private providers (verify current status under Kenya's digital health framework).

12. Payments and wallet

One checkout handles M-Pesa, card, insurance and wallet credit in any combination, and every sale produces an eTIMS-compliant receipt. The app should not hold stored-value customer money unless

counsel confirms the licensing position; until then, the "wallet" is loyalty credit and refunds, not a deposit account.

Payment methods

- M-Pesa STK push (Safaricom Daraja API) as the default; Paybill / Till fallback with reference matching.
- Airtel Money if volume justifies it.
- Cards through a PCI-DSS-compliant payment gateway; the app never stores card numbers.
- Split tender: insurance pays covered portion, user pays co-pay by M-Pesa, wallet credit applied first if chosen.

Wallet (loyalty credit)

- Earned from challenges, referrals, membership rewards and refunds.
- Family members can share one wallet under the account holder.
- Spend rules configurable (for example credit cannot be used on certain restricted items).

Health savings and financing (future, subject to regulatory review)

- Goal-based health savings with a licensed partner.
- Buy-now-pay-later or instalments for large purchases and chronic bundles through a licensed credit partner.

Receipts and tax

- Every transaction generates an eTIMS invoice via the existing Business Central integration.
- Refunds and reversals create matching credit notes.

13. Technical architecture and integrations

The recommended architecture is a modular backend with an API gateway, a FHIR-based health data store kept separate from commerce data, and Business Central as the system of record for inventory, pricing and finance. Specific frameworks and cloud provider are left to the build vendor to propose, subject to the data-residency requirement in section 14.

flowchart LR

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A[Android / iOS / PWA] --> G[API gateway]
W[WhatsApp / USSD] --> G
CC[Clinician & ops web consoles] --> G
G --> S1[Commerce services]
G --> S2[Care services]
G --> S3[Identity & consent]
S1 --> BC[Business Central]
S2 --> F[FHIR health store]
S3 --> F
S1 --> P[Payments adapter]
S2 --> I[Payer adapter]
```

Clients

- Android first (native or cross-platform, vendor to justify); iOS parity; PWA for low-commitment users.

- Target app size and minimum Android version to be set after device research on the actual user base.
- Offline: cached catalogue, record summary, medication list and reminders work without a connection; actions queue and sync.

Core integrations

System	Purpose	Notes
Microsoft Dynamics 365 Business Central	Inventory, pricing, orders, finance	Existing Xana RMS; reuse existing M-Pesa and eTIMS integration
KRA eTIMS	Tax invoices	Via Business Central integration
Safaricom Daraja	M-Pesa payments	STK push, C2B, B2C refunds
SHA and insurer platforms	Eligibility, pre-auth, claims	Interfaces to be confirmed (section 7)
Partner lab LIS	Orders and results	FHIR preferred, portal fallback
Video / voice SDK	Teleconsults	Vendor to propose; must support low-bandwidth fallback
Maps and routing	Store locator, delivery	Vendor to propose
SMS, WhatsApp Business API, USSD gateway	Notifications and alternative channels	Via licensed aggregators
Health Connect / HealthKit	Wellness data	Official SDKs only

Engineering standards

- Every service exposes documented, versioned APIs.
- Feature flags for per-module and per-region rollout.
- Event bus for cross-module events (for example "prescription approved", "lab result released").

14. Regulatory, privacy and clinical safety

The app handles health data, prescriptions, clinical services and payments, so compliance is a build requirement, not a later review. The list below is my understanding of the relevant Kenyan framework; I am not a lawyer, and every item must be confirmed by Kenyan counsel before design sign-off.

Area	Likely authority or law (verify)	Build implication
Personal and health data	Data Protection Act 2019; Office of the Data Protection Commissioner (ODPC)	Registration as controller/processor, DPIA before launch, consent records, breach notification process

Area	Likely authority or law (verify)	Build implication
Data localisation	Data protection regulations on processing health data outside Kenya	Hosting location decided after legal advice; design for in-country hosting
Digital health	Digital Health Act 2023 and related regulations	Confirm obligations for health information systems and telehealth providers
Pharmacy and medicines	Pharmacy and Poisons Board (PPB)	Premises and pharmacist licences shown; online pharmacy and delivery rules; controlled-drug restrictions
Clinicians	Kenya Medical Practitioners and Dentists Council (KMPDC), Clinical Officers Council, Nursing Council	Licence verification at onboarding and on renewal; telemedicine guidelines
Labs	Kenya Medical Laboratory Technicians and Technologists Board	Partner lab licence verification
Insurance	SHA; Insurance Regulatory Authority for private cover	Accreditation, claims rules, no selling insurance without licence
Payments and credit	Central Bank of Kenya	No stored-value wallet or lending without a licensed partner
Tax	KRA eTIMS	Every sale invoiced

Clinical safety requirements

- A named clinical lead signs off every clinical rule, triage flow, content item and self-order test list.
- Clinical incident reporting built into clinician and pharmacist consoles.
- Drug interaction and allergy checking from a licensed, maintained drug database (source to be selected and licensed).
- The app shows clear emergency guidance and never delays emergency escalation behind payment.
- Any AI features (symptom checker, chatbot) are decision support only, clearly labelled, logged, and clinically validated before release.

15. Non-functional requirements

The targets below are proposed starting points for the build vendor to validate and commit to in their proposal; none are based on measured Xana data yet.

Area	Proposed requirement
Availability	99.9% monthly for commerce and record services; clinical consult service 99.9% during operating hours
Performance	Home screen usable within 3 s on a mid-range Android over 3G
Security	Encryption in transit (TLS 1.2+) and at rest; field-level encryption for clinical data; MFA for all staff consoles; annual external penetration test
Access control	Role-based access; least privilege; break-glass access logged and reviewed
Audit	Immutable audit log for record access, prescriptions, dispensing and payments
Scalability	Horizontal scaling; load-tested to 10x launch volume
Accessibility	WCAG 2.1 AA; large-text mode; audio content for low-literacy users
Localisation	Swahili and English at launch; all strings externalised
Backup and recovery	Daily backups, point-in-time recovery for clinical data; RPO 15 min, RTO 4 h
Observability	Centralised logs, metrics and alerting; per-module dashboards

16. KPIs and analytics

The build must instrument these metrics from launch; targets are for management to set once baseline data exists.

Area	Metric
Growth	Monthly active users; households active; new members
Engagement	Weekly active rate; modules used per user per

Area	Metric
	month; wellness logging frequency
Pharmacy	Orders per user; refill on-time rate; Rx approval time; delivery time; order accuracy
Care	Consult wait time; consults per member; follow-up completion; patient rating
Labs	Booking-to-result time; abnormal result review time
Insurance	Claim approval rate; claim payment time; rejected-claim reasons
Wholesale	Basket size; business account repeat rate; group-buy completion rate
Membership	Conversion free→paid; churn; member savings vs fee
Clinical quality	Adherence rate for chronic patients; clinical incidents; escalations
Unit economics	Contribution margin per order and per member; CAC; LTV

Analytics events use a single shared schema across modules. Health data never flows into marketing analytics tools; product analytics use pseudonymous IDs only.

17. Open questions and items to verify

These must be answered before the vendor can give a firm estimate. Items marked "verify" are facts I could not confirm and should be checked with the named authority or partner.

Business decisions

- ☐ Clinical model: employed clinicians, contracted network, or partnership with an existing telehealth provider?
- ☐ Clinics: Xana-owned, in-store clinic rooms, partner clinics, or a mix?
- ☐ Lab partners: which labs, and exclusive or multi-lab?
- ☐ Wholesale: can non-members buy? Will Wholesale supply licensed businesses with pharmaceuticals, and under which licence?
- ☐ Membership pricing and benefit quotas per tier.
- ☐ Languages beyond Swahili and English.
- ☐ Geographic launch order (Nairobi first, then which towns?).
- ☐ Brand: "Xana Plus" as app name and membership name, or separate names?
- ☐ Build vendor vs in-house team, and who owns the product roadmap.

Items to verify

- ☐ SHA's current technical interfaces and accreditation rules for pharmacies, telehealth and labs (with SHA).
- ☐ Which private insurers use which claims platforms, and access terms (with insurers and platform providers).
- ☐ PPB rules on online pharmacy sales, delivery and restricted medicines (with PPB and counsel).
- ☐ Telemedicine guidelines and cross-cadre prescribing rules (with KMPDC and counsel).
- ☐ Data localisation requirements for health data and ODPC registration category (with counsel).
- ☐ Obligations under the Digital Health Act 2023 and any national health information exchange access (with Ministry of Health and counsel).
- ☐ Whether a national e-prescription standard exists that external prescriptions can use.
- ☐ Safaricom recurring-payment options available to Xana for membership billing.
- ☐ Stored-value wallet and credit licensing position (with counsel and CBK guidance).
- ☐ Licensed sources for the drug interaction database and Kenyan food composition data.
- ☐ Afyanalytics data-sharing: legal basis, de-identification standard and agreement terms (with counsel and DPO).

Added with section 10A

- ☐ Legal structure for medical harambee: Xana as collection agent or a licensed partner (with counsel).
- ☐ Diaspora payment partners and cross-border payment licensing (with counsel).
- ☐ Faith-layer content partners and review process across traditions.
- ☐ Verified data sources for air quality, water safety and disease-season alerts in Kenya.
- ☐ Whether insurers or employers will fund healthy-behaviour rewards, and on what terms.
- ☐ Mini-program partner SDK scope and vetting process.